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REPORTING · Issue 008, The biotech desk

Subnet Magazine Interviews Metanova Labs' Micaela Bazo and Dr. Pedro Penna

The CEO and CSO of Metanova Labs on building the first decentralized drug-screening platform in the world, and why a 65-billion-molecule database belongs on Bittensor.

⊕ 2025-10-02 · filed by Subnet Magazine

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discovery with NOVA: the first decentralized virtual screening platform in the world. Also known as RBittensor Subnet 68, NOVA's Terms of Service | Privacy Policy miners compete to identify the most promising small molecules from Accessibility | Ads info | More

a database of 65 billion the fastest. They're overcoming the immense computational challenge of running virtual drug screening using this decentralized AI network. @metanova_labs is driven by an R&D thesis that modulating key behaviors can have population-scale impacts

for health, wellbeing, and longevity.

@michaelabazo is the CEO of @metanova_labs, a crypto-native biotech company pioneering decentralized virtual drug screening and novel therapeutics.

Originally from Lima, Peru, she has a background in corporate strategy, systems design, master planning, and science-led innovation. Prior to co-founding Metanova, she worked as a Strategy Partner for RGoogle, where she helped translate emerging ecology and climate science into design and development strategies. She has led new program areas across frontier domains, developing performance-based standards, managing interdisciplinary teams, and aligning large public and private stakeholders around shared innovation goals.

Her work in ecology and resilience planning deeply shaped how she approaches NOVA, seeing both ecosystems and economies as adaptive systems that evolve through incentives, feedback loops, cooperation, and competition.

Metanova was born from both a scientific and personal conviction. She became fascinated with the possibility of developing drugs that help people break out of maladaptive behavioral loops associated with disease, suffering, and shortened lifespans.

A crypto investor since 2011, Micaela believes decentralized incentives have the power to reshape how scientific knowledge, technology, and society evolve. Her mission extends beyond drug discovery, to connect emerging scientific talent south of the equator with global capital markets and research ecosystems.

Pedro Penna (@pennapdr) is the Chief Science Officer and co-founder of @metanova_labs, where he leads scientific direction and model development across computational biology and integrated drug development.

Pedro holds a BSc in Science and Technology and a PhD in Biosystems from RUFABC (Brazil), where his research focused on bioinformatics, molecular evolution, and drug design. He also conducted studies in neurobiology at the Center for Neuroscience

and Cell Biology at the University of Coimbra in Portugal.

Before Metanova, Pedro founded and served as General Director at a biotech company specializing in computational approaches to therapeutic discovery. His work bridges molecular modeling and pharmacology, combining in silico hit discovery with experimental validation pipelines. He has a proven track record identifying promising hit and lead compounds and collaborating with world-class CROs on chemical synthesis and biological assays.

At Metanova, Pedro's vision is to merge decentralized AI with drug development, by leveraging RBittensor's adversarial incentivized framework to get to the AI models that can fulfill the potential for true disruption of pharmaceutical development.

SM: You share that @metanova_labs is the first "decentralized drug screening company". Can you share what that is, and what you are hoping to accomplish at a very high level, for those not only new to crypto, but to Bittensor, and the pharmaceutical industry.

Micaela Bazo (MB): Yes, I'm very proud to say that J]Bittensor Subnet 68, also known as NOVA, is the first decentralized drug screening platform in the world. As far as I know, we'll hold on to that title.

There's actually a lot of innovation to unpack in that statement, both on the business side and the technology side. I'm sure we'll get into it over the course of this interview. the course of this interview.

But to take a step back, when you think about drug discovery or the development of new medicines, it's a really, really challenging process. It's very prone to failure and part of that is because bodies are very complex systems in and of themselves. We need to be very responsible about how we go about identifying and incrementally testing something that can ultimately be deemed safe and effective and beneficial to broad populations of people.

We're specifically focused on small molecule discovery today, and it's been an incremental progress where we know that one thing works for something, and then we tweak it, and we get something else. With the advent of AI, we have been able to really transform this process and have increasing capabilities of efficiently identifying new molecules or

compounds that can be used to develop therapeutics.

However, we believe that there are limitations that are in part due to the incentives of biotechnology and pharmaceutical companies have in the development of new drugs. Namely, their valuations depend on proprietary, both technology, data, assets. So the vast amounts of data that are necessary to make these novel identification of therapeutics is not incentivized by our current system. What we're doing differently with NOVA is that we built a decentralized platform on J]Bittensor, meaning that we are using tokenized incentives to coordinate and reward work. What that allows us to do is tap a network of cost-neutral compute and labor, we don't have to go through a hiring process. Really, the challenge for us is translating these problems into something that is permissionless and verifiable. So can we make this into a competition where we reward people based on merit and not just based on accolades and have more liquidity through this process to reward iteration early on and therefore incentivize greater data sharing and access to compute that we wouldn't have otherwise?

SM: I'd love to hear more about you, @michaelabazo, and you, @pennapdr, about the team at Metanova and then your other teammates as well. What do each of you bring in your experience and where are all of you located?

Pedro Penna (PP): My background is in computational drug development with experience from my PhD in Biosystems, an experience building in vitro screening platforms and using them to generate new molecules with high pharmacological potential.

MB: Amanda (@42salamandas) has a background in AI, both in machine learning and computer vision, did her Master's thesis on relationship with neuro-pharmacology, and had background working in, a startup that ran diagnostics of genetic conditions in children using facial and clinical data.

And we were really lucky to also meet Braden, who has a background in computer science and biology, and most importantly, and very strangely, a background in Jbittensor. So early on, we kind of describe Metanova a background in Jbittensor. So early on, we kind of describe Metanova as spanning these verticals of AI, crypto, and bio. And we're a founding

team of four, and we each represent this pillar.

I have, as you earlier mentioned at the beginning of the interview, a background more in strategic planning on the corporate side, large scale systems from a variety of perspectives, science-led innovation, and so I'd say I'm more of the coordinator and the "dot-conconnector", but also, first learned about Bitcoin in 2011 and Bittensor in 2021 and was really excited about what that could mean in this context and how to bring those two worlds together. We've built this "Frankenstein" that's never been seen before, but I think we have increasingly a sense of conviction that we can help tackle some of the deficiencies that are currently plaguing the biotech business with this new model.

SM: You've shared in the past that Metanova is disrupting the pharmaceutical industry, which has traditionally been "pathology- centric", reactionary to diseases and side effects of those diseases that are affecting people at a large enough scale to be marketable. Can you share your R&D thesis on why you chose to go the route of behavioural drugs that promote wellness and longevity instead of disease and pathology?

MB: The reason why I think that it's so centred on pathology is because that's ultimately how we evaluate whether or not a drug will be approved. So the FDA is basically assessing whether your candidate is going to be safe and effective for "x" condition. That has a whole series of effects on the way that new drugs are thought about, funded, backed, or not. And I don't think that's necessarily the best way to think about what constitutes health and well-being. I think it made sense as a way to operationalize the approval process, and it's still a really important consideration when you think about what are going to be the commercial opportunities of such a risky, multimillion dollar bet as on average has been drug development in general.

But I think for us, especially being in step one, we're in an early stage. We're doing virtual screening. We are the very beginning of this long development cycle. I think it really begs the question from a first principles perspective, what leads to health and wellbeing? What are these upstream determinants of how happy you are, how likely you are to contract to the disease? Infectious diseases are a whole other piece, but even then, there are certain things that may compromise your immune system.

So we identified a gap in thinking about mind and body together. Those have also been severed and focus on this metaprogramming concept or behavioural concept and recognizing that there are key behaviours that are foundational to health and life outcomes, whether it's sleep, or focus, or eating, or social connection.

And I think in large part, we've talked about this before, but that helps explain the magnitude of effects that we're seeing with GLP-1 agonists, like Ozempic and Mounjaro. We don't necessarily understand how all of these things work, but very commonly in the pharma industry too, you

release a drug for one condition and then over time you find off-label uses for it. We developed Lamictal for convulsion and seizure activity, and it ended up being one of the most important treatments for bipolar disorder. So similarly, in our ideation of "what are the gaps in medical innovation?" and "how do we pick strategic drug targets that ultimately will give us the flexibility to respond to the data that we get on our specific compounds?" and therefore seek approval for specific conditions that fall in this regulatory framework, that's pathology- centric, we're trying to think about what will have the most impact on society. And what where are these upstream axes and where do they intersect?

SM: I would think that traditionally, what we call "big pharma", those that research, manufacture, sell, prescribe the drugs, the industries that have helped cause those diseases (like the food industry), have made billions if not trillions off of this pathological model. Do you anticipate disrupting this industry will be difficult?

PP: The pharma industry is really hard. Developing a drug requires large investments. The friction rate is very, very high, meaning the failure rate is pretty relevant at each one of these steps that must be done for you to get a drug to be approved by regulatory agencies. So for sure, this is no easy task, but we really think that the value proposition that we're bringing here makes total sense in terms of optimizing for increasing the odds of success of having the drugs that are going to be doing what we want them to do and not doing what they don't want them to do.

MB: I think when we talk about "disrupt", the beauty about incentive mechanism design, is that you can align with your friends and enemies for a common direction. Not to peg anybody as the good guy or the bad guy. That's not my job.

Our mission is bigger than that. I think part of the reason why they need to make so much money is because the failure rate is so high. So one

drug is paying for nine failed attempts. That's another way to think about the economics of drug development here. Is this going to be a hard thing? Anything, anything that's ambitious is going to be hard, but I feel like we have an unnatural opportunity of making a dent in this problem. And first mover advantage that puts us in a really good position to attract their interests.

We've specifically designed a platform that can support not just our own R&D interests, but also that of others. And that's something that's really, really rare in the current biotech market and that it's only really possible because we're using decentralized AI and tokenized incentives. We are witnessing a moment where other biotech startups are having a really hard time backing very promising pipelines. I'm optimistic because we're disrupting an industry technologically and economically in a way that makes room for all of them to align with us and not necessarily have the incentive to take us down, but rather make use of the thing that we are creating. creating.

SM: For many not familiar with the pharmaceutical industry, after COVID-19, and with the exponential growth of the AI industry, you would assume the pharmaceutical industry would not only readily adopt the AI approach in the same way Metanova has, but that VCs would readily fund these projects. Can you speak to why this is not the trend?

PP: Whenever we talk about disruption in the pharma industry, we are talking about a very hard-to-tackle industry, because as Micaela put it, bodies are really complex, biological systems are really complex, those things usually take time. If we look, for example, how is the current state of the drugs that are approved by the pharmaceutical industry? You can see that 60 to 70% of everything that is approved came from academic labs or small companies that are called virtual biotech. Usually the two align together to create a new interesting asset. This was only possible because there was the advent of CROs or contracted research organizations that are companies focussed on offering, as a service, chemical synthesis or wet lab assays, or in vivo assays. And it took time for it to really consolidate and become the basis for this kind of disruption in the speed of innovation. I think with AI, I think something that is similar, we have a lot of really interesting AI approaches and really interesting companies developing drugs with AI. But it takes time to get us drugs to the final line. So I think we're about to see a lot of very interesting things with dTAO.

SM: What kinds of diseases have you screened for so far? How do you choose what protein targets to work towards? Is it in terms of marketability of these particular protein targets and the drugs that target them?

MB: We launched with an emphasis on reward and learning mechanisms, looking for the first principles that we think will allow us to find entry points to develop novel therapeutics for a wide variety of conditions.

PP: Our first set of targets were monoamine transporters, and then we moved into more novel epigenetic regulators, that that we're very excited about. Some, to say, are more validated than others in terms of how many drugs have been approved for each in the conditions that we would be most interested in.

Considering all that we're interested about, on modulating behaviour and how we interact with the world and all the consequences of that, we chose two key protein families. The monoamine transporters, that covers the serotonin transporter, the dopamine transporter, the norepinephrine transporter, and histone deacetylases. And it was a strategic and well-thought-out choice, like every target selection during this process needs to be. In our case, those two families check all the boxes that are looking for, so it could be safe that investing resources, time and focus would lead us to something that has a very high likelihood of being useful for our goal here. Those boxes, to say a few of likelihood of being useful for our goal here. Those boxes, to say a few of them:

- Strong genetic validation in association with key behaviours, and conditions that are really important
- A lot of scientific papers in the literature evaluating the effect of modulating those targets in cell lines, in animals and in humans
- A lot of very clear paths for therapeutic indications in many different kinds of indications in the case of the two families

For example, for the monoamine transporters, we have some very interesting indications, usually in the neurological, psychiatric realm, like ADHD. But also expanding to potential in Parkinsons, chronic pain, addiction, they are really well established in terms of addressing a lot of

depression symptoms. So they are pretty broad. They control how we communicate with this specific class of neurotransmitters, the monoamines. So they have a really, really relevant impact in terms of mood and how we interact and perceive the world.

The histone deacetylases are even more versatile, epigenetic regulators, so they control how different genes can turn on and off. They hold a lot of potential, for depression, but inflammatory conditions, Alzheimers, a lot of discussion in terms of their use for aging and anti-aging approaches. Interestingly, we have FDA approved HDAC modulators and also modulators for several of those transporters that I just cited, the monoamine transporters, for some of those indications, so this is even more validation about those targets.

But on top of that, there's a lot of room for improvement. So a lot of side effects and a lot of specific properties for the already approved molecules that it could be so much better if they were working a different way. In some cases, it can be a different selectivity profile. It would be targeting and binding to a field, but not all of the deacetylases, for example, or it would be blocking with higher intensity, one of those channels, one of those transporters.

So all of those things were considered for us to pick those families of targets. In a very concise way, we're looking for things with the right level of validation, in terms of being useful to really not only treat but modulate how we're interacting with the work, but also with enough space for optimization. So we could really make a difference here.

MB: It's been really exciting in this long journey of trying to tackle this big problem is the new research that comes out as we're working on these problems. We literally were running serotonin as our target of the week for the NOVA competitions, and they had just published this new paper on its promise for immunotherapy and tumour growth suppression. That's a completely new area that we don't necessarily associate serotonin as a useful drug target for. But I think that's why we're trying to look at the intersection here, because it will have the most potential, not just immediately, but also over time. To your earlier point of our interest in population-scale impact and thinking beyond just one pathology. SM: Micaela, you shared a very touching post earlier this month,

where you shared that the launch of Metanova aligned really closely with the passing of your father. Can you share that experience, and what inspired you personally, as CEO, to focus on drug discovery?

Micaela @micaoabazo · Oct 4, 2025 when i realized the incentives for developing better drugs are broken, i decided to do something and co-founded @metanova_oabs.

my father's passing only made this more personal.

small molecules are one of our greatest medical breakthroughs yet progress has plateaued:

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MB: I think like any founder, it's really hard to really draw a line in the sand and be like, "this was the exact starting point", because it takes so much time and energy to launch a company that you better be obsessed about solving the problem at hand.

I've always been really drawn to biology and chemistry. It's what I thought I wanted to study in college but didn't. I pivoted many times, and I had a growing interest in how you can support people's sense of agency and adaptability, which led me to a fascination in neuroplasticity and how you help people break out of negative behavioural patterns. Like really understanding more and more this "you problem" thing.

And I think with my dad, I think he suffered from depression, I suffer from depression. And I saw that it had a correlation with his lifestyle, which, all hypotheses led to his cancer diagnosis.

But the experience of first-hand and third party and overall I think, a reflection of what I think is associated with capping our potential and leading to much suffering. I also saw the power of the mind, so when he was diagnosed and treated with it, I obviously dove into the science and was reading the really, really, really bad prospects that are associated with the Stage 2 pancreatic cancer, and I simultaneously didn't want to burden him with that knowledge. Because ultimately, we were just trying to understand, what are the odds all the time? He assured everybody that he was cured, he had a surgery, the chemo was preventative, even though that's not what the literature would suggest. And the fact that he ended up being such an outlier in his case, I think, really confirmed this interest in my thesis in the power of "brainwashing yourself". That's another way of really calling neuroplasticity, but it's like, how do you, all of a sudden, take control over your actions, over your belief systems, over your thought patterns, because I do think they have a really meaningful impact on your health and wellbeing.

This led me to do a lot of research, and I talked to, I think, 70 different people looking for a technical founder until eventually I found Pedro and then Amanda and then Braden, and I'm really happy that together we're working on this project and building something that's never been done before, because I think there is a clear need to think about health from a before, because I think there is a clear need to think about health from a different perspective in the development of novel therapeutics. And I think we can meet with the trad sector. We don't have to be in parallel paths. I think, for me, that's what also inspires me to keep going every day.

SM: For those that are new to or unfamiliar with the process of Metanova's role in virtual screening and selection of small molecules, to ultimately in vitro testing, development and approval, can you take us through, in simple terms, the full process?

PP: Drug development takes 10 to 15 years, it costs billions of dollars. Most failures happen because we pick the wrong molecules early on and keep developing them. So it is like building a house, but you really, really need a good foundation for it to work. If you think about it traditionally, early stage in drug development, so in the early stage virtual screening, we are testing millions to millions of molecules. It takes months, it surmises a lot of promising candidates.

This matters a lot because it's very early screening cascades, they are super relevant for the entire pipeline. If you can find better starting molecules, you are then avoiding pretty costly late stage failures. If you can screen a lot faster, you can explore a lot more, the chemical space and potential diseases that you can then tackle.

So better computational predictions, they mean less trial and error in the lab, which, by the way, usually the process goes like that. You do the computational screening, after that, you have synthesis of compounds, you have them assayed in a wet lab, then you go for animal assays and then it goes for human assays. There are key points in all of this development where you need to check a few boxes so you can go on. And that is why it takes so long and it is so, so, so expensive. So our focus here is truly how we can reduce the amount of synthesized compounds and reduce the amounts of tests. And ideally get it right from the very, very beginning. This is the core of how we are looking at this process.

SM: There was a recent RDSVfund interview with RRidges_AI, where an audience member had posed a question about “the difficulty of

stopping other companies from using Ridges' open source agent code”. The community member stated, “Although the open source method is great for pushing the boundaries, I think it could also be difficult to defend.” RRidges_AI' response was that their model is one in which the agents are getting better every day, so by the next day, the model that gets “stolen” is obsolete.

DSV @dsvfund · Oct 6, 2025 Bittensor Revenue Search 34: SN62 Ridges Special AMA
x.com/i/broadcasts/1...

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You shared that the next step for METANOVA is to start getting miners to submit their code, and of course this IP is really important. miners to submit their code, and of course this IP is really important. What would prevent a pharmaceutical company from adopting this code and running their own version of your models? And are the iterations of your miners getting better at a fast enough rate that this is not a problem for METANOVA?

MB: So we're really proud to announce that actually, that phase is now available for testing, so we will be very soon launching a new (formerly known as a sub-subnet), a new incentive mechanism, so that we can continue to both receive submissions of molecules, and then, as you were saying, algorithms for search approaches.

METANOVA @metanova_oabs · Oct 14, 2025 #bittensor #SN68 is evolving

introducing NOVA Blueprint: an open algorithmic challenge to innovate faster, smarter molecular search methods for ultra-large datasets that can plug into any model for any property.

this will be released as a new incentive layer inside the subnet

4 24 98 28K

There's many different kinds of things to monetize here. And I think we made a strategic decision to push on the open source innovation of the search algorithms, and they can be used to accelerate. They can plug into any model that is predicting for different kinds of properties. What we're innovating here are search approaches. So in that way, do I think that any company would be able to monopolize the entire market? No.

And at the same time it could also be an opportunity for a breakout. Imagine that one of these giant companies validates code that's being submitted by one of our miners. We will be tracking their improvement over time. We currently don't have any results to share about that specific part of the challenge. But also we have a moat when it comes to the larger value items that will be developed by, or as a consequence of the subnet.

SM: Once you have a potential molecule, what patents, or IP protection are you afforded? What does that look like within the pharmaceutical industry? pharmaceutical industry?

PP: So getting back to these phases that we were just talking about, the pharmaceutical industry is structured around critical inflection points, where IP transforms into commercial assets. When you identify a promising molecule through virtual screening, this is the beginning of a patent cascade after you have it synthesized, tested in cell models and animal models. After you do it, one can, for example, file composition of matter patents that cover the molecule itself. Those are pretty strong, they will give you around 20 years of protection and exclusivity to explore this specific IP.

As development goes by, every breakthrough can create new IP opportunities. So in some places of the world, you have methods of use patents. So you discover specific therapeutic applications for the molecules that are covered in composition of matter. You can also have formulation patents, when you optimize how it is going to be delivered, you can have process patents for how it is going to be produced. The real value here emerges at those validation milestones. It is very long, it is very expensive.

So successful individual testing, animal efficacy data, Phase 1 clinical safety results, Phase 2 efficacy results. Each one of those milestones, they create a significant derisking event that then, in a very relevant way, increases how much this asset is worth.

That is why the pharma industry operates as this massive licensing ecosystem. A startup might discover and patent a molecule, then license it to a mid-sized company after preclinical validation or early clinical validation, then partners with big pharma after Phase 2 clinical trials.

If one of those inflection points from the computational test to the clinical proof of concept really represent both potential patents of all

innovation and a potential dealpoint during this whole development.

So the earlier you can computationally guide the validation and patenting of the promising molecules, the more shots you're going to have for those licensing events. That is how we see NOVA's accelerating this whole process to create more IP earlier and enable companies to build stronger patent portfolios, being able to negotiate from good positions and optimize the whole process as a whole.

SM: You mentioned on Metanova's X account that after launching the new incentive mechanism on testnet, you will begin synthesis and in vitro testing early next year. What does that look like, do you have partnerships with in vitro labs? What role does wet-lab validation play in your ongoing assessments? Can you share what CROs you are looking to work with?

PP: So from my post-doctoral work, I interacted with many labs, CROs, focused on chemical synthesis, focused on performing wet lab assays. But we know that computational predictions only really fulfil their full potential as that you have wet lab validations. Getting them back quickly potential as that you have wet lab validations. Getting them back quickly to patents, you really can only submit a strong patent for this kind of innovation if you have the molecule backed by this wet lab data, that is why we expect to be starting the synthesis and testing of compounds in Q1 of next year.

So we're in discussions with multiple CROs and our plan is to start simple. So start with 50 to 100 high confidence molecules, validate the computational models against real data, scale really fast, based on the hit rate and picking what is really working and being validated from those new approaches and new predictions. So ideally, we want to have it started between January and February in terms of starting chemical synthesis, initial in vitro data by Q2.

And it is important to point out that it is not just about proving that the platform works. But it is about transforming all that is being done here into defensible IP, that the pharma companies will also see as valuable and will, reduce substantially, the power for us to be interacting and showing the potential of using this kind of crypto-based decentralized system in accelerating this kind of innovation.

Also important to add, I work with multiple of those CROs. The moment we are happy with the code submissions that are being done and being

used, we can then put it through here knowing exactly the level of quality that we will have in terms of data coming from those.

MB: And I think for us, it's really important to both use this as an opportunity to advance potential drug candidates down the road, like push our own R&D, but also build our platform.

And I think we made the strategic decision to first launch this code because we saw how successful]Ridges_AI had been very, very quickly, potentially. out-creating or beating one of the largest AI giants in the world. So we are being very strategic about how we use this investment, but we're also looking for opportunities in general to build the platform as a whole.

So the supporters of NOVA, this is what you're helping with. You're helping us not only push our own R&D interests, but build up a platform at a time where very, very few companies can actually afford to do so. Many biotech startups that are currently in operation were founded with the intent of pursuing a platform, but don't have the advantages that we believe that we have with decentralized systems to really maintain that and have scaled down to singular bets. So I think we are more economically resilient and in a position to push a different kind of technology thanks to the adversarial challenges that we have.

SM: An interesting facet of the project is what you've called "geographical arbitrage", where you utilize the high quality and quantity of research coming out of places like South America, South East Asia at competitive pricing. Why do you think there is a disparity in how scientific discoveries from the Global South are seen in the Global North? Do you believe working with certain CROs, certain Global North? Do you believe working with certain CROs, certain companies will help or inhibit Metanova downstream when you get to

FDA approval one day?

MB: We're very mindful about how we are crafting our strategy. And I think actually the fact that both of us are not originally from the U.S. is actually an advantage, because biotech has been extremely concentrated in the U.S. Part of that is just that funders felt more comfortable funding people that had gone to universities that had a certain name and that they could meet in person.

And, in fact, Pedro and I had some original conversations with some investors that were basically saying, "Oh, you should just pretend like

you live in the U.S", which I have a hard time doing. Point being is there's a huge opportunity here for geographic arbitrage, not just in the virtual screening process, but in the preclinical and even clinical development. And in crafting that strategy, we've already thought about, what are the most strategic locations to do this? With a goal of still getting FDA approval. And the FDA has signed some interesting agreements with governments from around the world that actually make it possible for you to leverage reputable labs that are operating at a fraction of the cost and especially when it comes to clinical trials, are in places that have very diverse populations, which is an important thing when trying to determine whether or not a drug would be successful for different types of people.

So decentralization is not something that we just take seriously from the virtual screening perspective. It's, I think, something that we will continue to push in every step of the research and development and I think like a huge opportunity that most people maybe don't have the level of experience to trust necessarily. So one of our missions, we're pretty ambitious, but I think for me personally, one of my motivations in connecting with co-founders that were outside of the U.S was building that bridge and knowing firsthand that there's so much talent that doesn't end up crossing the border, but that has so much to contribute and that needs to be tapped in this next innovation cycle.

SM: And as a quick followup, do you have a gauge of where your miners and validators are located in the world?

MB: They're everywhere. They really are. We met some of them in person. We attended this conference called Proof of Talk, put on by JXVentures_Fund, which had a JBitsensor track back in June and had the opportunity of meeting some of them, based in the USA, Latin America, Europe, Asia. Bitsensor is really impressive in its global reach. And our miners had all different kinds of backgrounds, like very talented people, really smart, didn't necessarily have a background in this specific problem, which I think was confirmation that we did a good job of designing the problem because we were trying to remove the need for domain expertise.

However, the incentives are really high. I think our miners after the burn are making about \$6,800 a day, like \$2.5 million a year. And they very are making about \$6,800 a day, like \$2.5 million a year. And they very quickly learned the concepts that were maybe outside of their original

fieldhouse, but that made them increasingly competitive, allowing them to top those rewards.

Just gauging from the level of discussion that was happening on our Discord channel at times, it was very clear that some people had brushed up on some foundational concepts.

SM: Metanova has been working to build off of the SOTA opensource models to further refine your future drug affinity predictions. Recent updates include integrating Boltz-2 for structure-based predictions and adding Suzuki-Miyaura coupling to your datasets. Do you anticipate adding more models to your deck?

METANOVA @metanova_oabs · Sep 16, 2025 NOVA update: Suzuki–Miyaura coupling added to our combinatorial reaction dataset

+17B new possibilities unlocked for miners (+16,974,045,180 unique molecules, to be exact)

this Nobel-prize winning chemistry is one of the cornerstones of modern drug development.

why?

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PP: Yeah. Not only more models, but also more chemical reactions, more billions of potentially synthesizable molecules, and not only more models in terms of affinity, but also more models that are going to be predicting other different relevant pharmacological, biological features for the molecules, the solubility of the molecules, their half lives, potential toxicity issues, creating really an end-to-end prediction platform for development, to help those orthogonal models, like PSCHIC, finetune models in Boltz-2 to really increase the confidence you can have in terms of the affinity, but also predicting a lot of other, very, very relevant pharmacological properties that we need to take into consideration when developing a drug.

SM: In the past, miners have had objections to Metanova's choice to burn emissions to strengthen the overall price of the alpha token. I believe it was correlated with the decline of your number of miners early on in your journey, from 250+ to less than 100 now. Can you speak to that dynamic, has your relationship with miners gotten better? What is METANOVA doing to incentivize more miners to the better? What is METANOVA doing to incentivize more miners to the

subnet?

Miners on MetaNOVA's dashboard - October 19, 2025

MB: I think, first of all, it's impossible to really know how many miners there are. It's something that we've tried to iterate about, like, how do we best capture that? How do we best track that? And, in fact, one of the things that you need to take into account when you're iterating on the incentive mechanism is: is this going to result in or incentivize miners to register more slots, essentially appearing as if there's more of them? Because there's a competitive advantage in just submitting answers in that way.

Yes, the miner burn was something we followed the lead of other subnets, and to be fair, in a way, we're responding to historical conditions, like the 18/41/41% was something that was said early on when there weren't even that many subnets. And I know it's something that they want to change actually pretty soon. We were ultimately making a decision that we thought would preserve our ability to offer our miner incentives, as well.

But, yes, of course, it's controversial. Like they have every incentive to push back on making less money. Like that just makes sense. And I don't know if there's a perfect objective answer as to what is the right

number, but we are very interested in attracting growing numbers of miners and working more closely with them.

I would like to think that we have a better relationship and in fact, they were really helpful when we were releasing Boltz-2 in identifying some gaps, like there were some translation that we needed to do to make sure that there were consistent results because there was a stable diffusion model and they were part of the testing.

I can't definitively tell you how many miners are participating because they might be testing strategies offline and not submitting them for the challenges. There's a lot of nuance here, where it's really impossible to tell you exactly how many people are trying or competing actively on NOVA's challenges, but we're hoping that with having multiple incentive mechanisms and with the introduction of this new open source, code-based competition that it will lower the barrier of entry and attract new mining groups. We're also always talking to diverse groups of people that are interested, potentially in either mining our subnet or in seeing if there are any synergy points for collaboration of the application of their

submissions as well. We're open to talking to any new people that are interested in taking on the NOVA challenge.

SM: The scientific community is largely unaware of the tremendous amount of work happening here. Have you, or do you plan to present your findings at the academic level at all, at conferences or through published papers?

PP: Yeah, at some point for sure, we have not yet done that, but I imagine with this new structure of incentive mechanism, this will also accelerate the time that we expect to have very interesting data created by all the community together in a way that can plug really well with the scientific community.

SM: Earlier this year, RYUMALabs announced a partnership with the University of Connecticut (RUConn), to launch RBittbridge - a new program that engages the academic institution and its students to build on Bittensor.

RZEUSubnet, SN18 posted that the team presented at the Amsterdam University of Applied Sciences demonstrating how weather forecasting benefits from a permissionless, decentralized system. This week, they shared, students will actually get to build ML temperature forecasting models as miners for ZEUS.

I think this model, or this approach, has enormous potential. Metanova could work with undergraduate or graduate level bioinformatics or molecular biotechnology students to engage with what you're doing at Metanova.

I would love to see a day where Metanova is working with a lab and students are miners, winning rewards and also getting academic credit for it. I would be curious to hear your thoughts on this model to engage new miners.

PP: I mean, that is like really interesting and I think the good thing of working in this permissionless system is that it actually might be happening already. At the beginning, we had miners posting on X from laboratories inside universities, pictures tagging NOVA.

So this might be happening already. Because like the background of the miners are truly, truly diverse. We can see that they have a very strong analytical, mathematical and data science background, but it is very, very varied.

So actually, if that is happening, getting in contact with us, that would be great too.

MB: Yeah. And I think JBarbarian7676, I think is 16, working on one of the most anticipated and well-regarded subnets. I think he's 16. He mined NOVA early on. We met in Paris, had some exciting chats. But yeah, I think there's definitely opportunities to collaborate with different yeah, I think there's definitely opportunities to collaborate with different university groups. Those are on the table.

SM: You've shared that the short-term revenue will come from provision of services (like drug-screening-as-a-service, model-training-as-a-service). Where are you in that process? What steps are you taking to attract pharmaceutical industry partners while maintaining the core thesis of decentralization and decentralized Science?

PP: So this is actually a very interesting thing because when talking to the industry, one very important point that was raised to us was, "Okay, but usually in many cases we're not even disclosing our targets. You're telling that we need to disclose our targets and anonymous miners are going to know the molecules. I'm not totally sure about that". Even though we see a very clear path on how this can work, very traditional players in the industry might want to be more careful, but direct feedback from them pointed to how this new incentive mechanism that will be running in parallel really, really can solve for that.

Because, for example, if very innovative new code is being generated and then we are turning that into something that really solves the needs of these specific partners, then this problem is not there anymore. So we are in very active discussions with multiple players, but something like that was not done yet. So we need to find the right way to plug.

MB: I think and I constantly talk about, what's the right balance between being pragmatic and being a visionary or just trying something new? And I think our company has always reflected this hybrid approach. decentralize the pieces that make sense. Keep the things that make sense, centralize. We're already trying to do something that's never been done before. And there will be a transitional period where we bring a sector that has no crypto exposure or growing exposure to crypto, maybe just as an investment in their treasury. It'll be an inevitable

convergence point. But as you're writing the script on the fly, you need to make some strategic decisions about where are the points where you're going to conform to the current system for the sake of collaboration, for the sake of adoption, and where you really gonna benefit from trying something differently or approaching it from a different perspective.

SM: There are a lot of big questions in science that could be solved using the Bittensor framework. For aspiring entrepreneurs in Decentralized Science, what advice would you give based on your experience building Metanova?

MB: My advice would be go big or go home. Honestly, truly. You need to be using the power of magical internet money and global decentralized networks to push something bold and new.

It is such an amazing opportunity. Don't waste it on Vitamin C. Really push innovation here and I think there's enough precedent that actually it's not that crazy of an idea. There's many success cases within the crypto universe and 10 years ago, nobody would have ever thought that some of these things would be so mainstream. So dream big because I some of these things would be so mainstream. So dream big because I think the future is yours to build and yours to make. And what could be

better than using a global financial mechanism to build an entirely new reality when the world needs it the most?

SM: Best case scenario, you get the model-of-all-models, and can choose the molecule best suited in all cases to solve one health problem, either all the way upstream or all the way downstream. What is the problem you would choose to solve?

PP: I would say for me, I think anxiety and depression, because those two come together, would be absolutely relevant to solve. They are really, really growing, and I think they have such a relevant impact. And in many cases, cap humanity growth, that so many other problems would be solved if we could solve this one really well.

MB: I think tagging along with that, and I think related, is connection. We know the U.S., in fact, went as far as calling loneliness an epidemic. And I think it is closely associated to anxiety, to depression, to your life expectancy. And I think a growing challenge and an increasingly digital world that seems very alienating and hostile and apparently here to replace us all together.

So I think if we can help people come together, if we can help combat loneliness and adapt to a completely different planet than what our species adapted to many, many years ago, then I think that will probably have the most like cascading effects.